Zirconium Copper CuZr – Luvata ZrK015

Typical applications:

- Resistance welding electrodes
- Spot welding adaptor shanks
- Commutators
- Other applications where high conductivity and good resistance against softening are needed

Alloy description

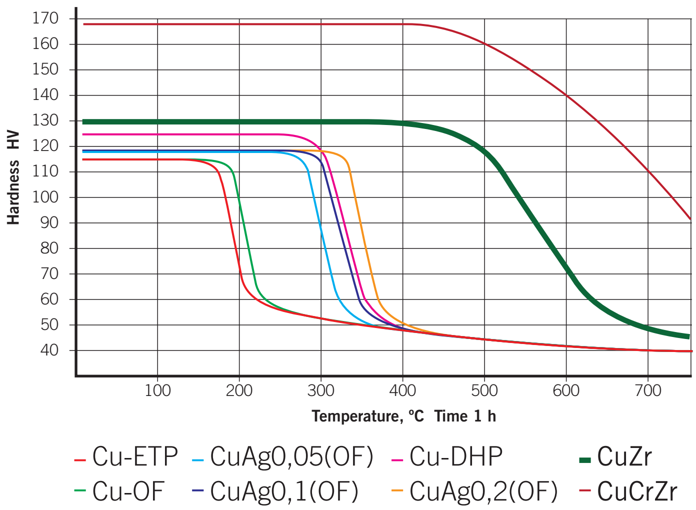
Luvata ZrK015 alloy is a precipitation hardening alloy for high temperature applications where material need to have a combination of high electrical and thermal conductivity and mechanical properties. Mechanical and electrical properties of zirconium copper are obtained through thermomechanical treatment which typically consists of following steps: solution annealing followed by rapid quenching to water bath, cold working, aging at 400-450°C and cold working to final dimensions. The final metallographical structure of zirconium copper consists of finely dispersed Cu_5Zr precipitates which develop during the aging treatment. Aging treatment is therefore essential to achieve high resistance against softening at elevated temperature and high electrical conductivity. ZrK015 alloy can be supplied as aged temper or without heat treatment.

Products / shapes:

Round rods, wire coils, rectangular bars and solid profiles in age-hardened temper.

Softening behaviour – resistance against softening:

Room temperature hardness is presented in the following figure as a function of annealing temperature. Material at hard or aged temper.



Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition %	EN – CEN/TS 13388:2008	ASTM / USA
ZrK015	Cu + Ag min 99,8 % Zr 0,1 – 0,2 %	CuZr / CW120C	CDA C15000

Physical properties:

Density kg/dm³	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,89	0,0000169	385	1080

Mechanical properties – round rods:

	Round dia < 25 mm	Round dia 25 < x < 50 mm	Round dia > 50 mm
Hardness HV	120 – 150 HV	120 – 140 HV	Approx. 120 HV
Tensile strength	350 – 450 N/mm²	350 – 430 N/mm²	Approx. 350 N/mm²
0,2% yield strength	300 – 400 N/mm²	300 – 360 N/mm²	Approx. 320 N/mm²
Elongation	8 – 25 %	10 – 25 %	10 - 25 %

Electrical and thermal properties – typical values:

Electrical conductivity	vol	% IACS *	approx 90,0
	mass	%IACS	approx 89,8
	MS/m		approx 52,0
Electrical resistivity	vol	Ω mm²/m	approx 0,019
	mass	Ω g/m²	approx 0,171
Thermal conductivity (20 °C)	W / Km		367

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Good	Good	NOT Recommended	NOT Recommended	NOT Recommended

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Data sheet

General description. Subject to changes or deviations

Copper Nickel Silicon Alloys - Luvata NK101, NK103 and NK203

Alloy description

CuNiSi alloys offer high temperature endurance to basic copper properties. Strongest CuNiSi version is CuNiSiCr, which has similar properties to beryllium containing copper alloys. Luvata CuNiSiCr is beryllium free alloy for medium high electrical and thermal conductivity, and it can replace harmful beryllium copper alloys. Intermediate strength is obtained with CuNiSiCrZr alloy.

Mechanical and electrical properties are obtained through thermomechanical treatment which typically consist of following steps: solution annealing followed by rapid quenching to water bath, optional cold working, aging at 400-500 °C and possible machining to final dimensions. The final metallographic structure of CuNiSi consists off finely dispersed Ni_2Si. When Cr or Zr is added also they precipitate as silicides. Adding chromium or zirconium to basic CuNiSi material is possible and that way strength can be added. Aging treatment is essential to obtain high mechanical properties and sufficient electrical conductivity.

All CuNiSi alloys are also available in solution treated state. In that state these alloys can be bent, straightened, or otherwise formed. After aging treatment only machining operations should be made. Instructions for all necessary heat treatments are available from us.

Typical applications:

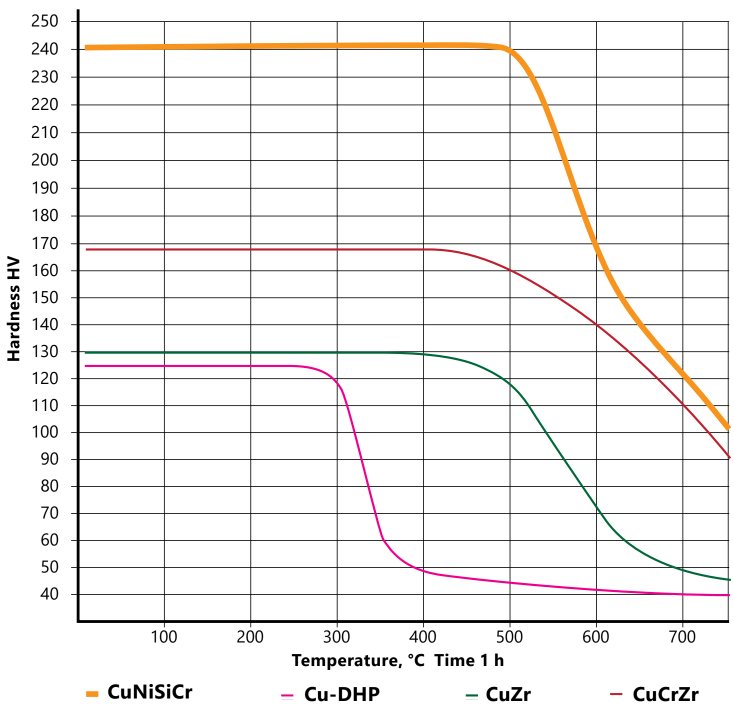
- Spot welding adaptor shanks
- Generator rotor wedge profiles
- Casting dies for non-ferrous metals
- Other applications where high strength material properties are needed.

Product / Shapes:

Extruded and drawn round rods and solid profiles. Various shapes as forgings or machined components.

Softening behaviour – resistance against softening:

Room temperature hardness is presented in the following figure as a function of annealing temperature. Material at hard or aged temper.



Partnerships with a Promise

Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition	EN - CEN/TS 13388:2008	ASTM / USA
NK101 CuNiSi	Ni 1,6 - 2,5 Si 0,4 - 0,8	CW111C	CDA 70260
NK103 CuNiSiCrZr	Ni 1,6 - 2,5 Si 0,5 - 0,8 Cr 0,05 - 0,15 Zr 0,15 - 0,25		
NK203 CuNiSiCr	Ni 1,8 - 3,0 Si 0,4 - 0,8 Cr 0,1 - 0,8	CW111C	CDA 18000

Physical Properties:

Density kg/dm^3	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,8-8,9	0.0000175	380	1020-1060

Mechanical properties – typical values:

	NK101	NK103	NK203
Hardness HV	130-240	180-220	220-250
Tensile strength	420-780	590-750	650-800
0,2% yield strength	380-620	>520	600-750
Elongation	9-16%	>5	9-15%

Electrical and thermal properties – typical values:

Electrical conductivity	vol	% IACS	38-51
	mass	% IACS	38-51
	MS/m		>22
Electrical resistivity	vol	Ω mm_2/m	0,033-0,044
	mass	Ω g/m_2	0,033-0,044
Thermal conductivity (20 °C)	W / Km		220

Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Good	Good	NOT recommended	NOT recommended	NOT recommended

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Data Sheet

GENERAL DESCRIPTION
– SUBJECT TO CHANGES OR DEVIATIONS

Chromium Zirconium Copper CuCrZr – Luvata KrK101

Alloy description

Luvata KrK101 alloy is a precipitation hardening alloy for high temperature applications where material need to have a combination of high electrical and thermal conductivity and mechanical properties. Mechanical and electrical properties of chromium zirconium copper are obtained through thermomechanical treatment which typically consists of following steps: solution annealing followed by rapid quenching to water bath, cold working, aging at 400-450°C and cold working to final dimensions. The final metallographical structure of zirconium copper consists of finely dispersed Cu_5Zr and pure chromium precipitates which develop during the aging treatment. Aging treatment is therefore essential to achieve high resistance against softening at elevated temperature and high electrical conductivity. KrK101 alloy can be supplied as aged temper or without heat treatment.

Typical applications:

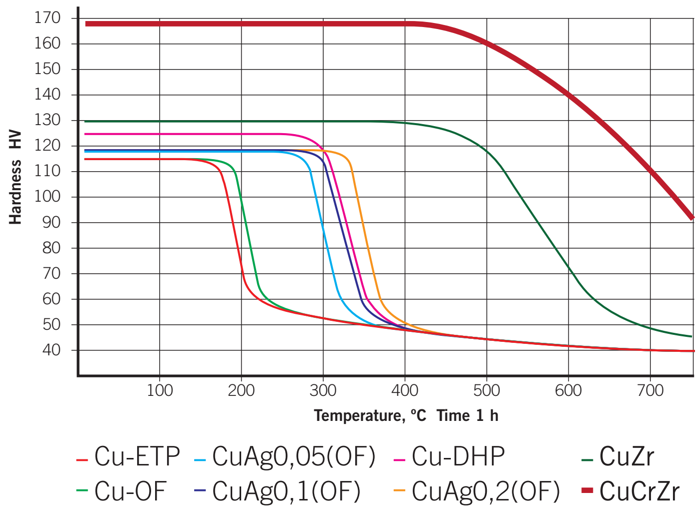
- Resistance welding electrodes
- Spot welding adaptor shanks
- Other applications where high conductivity and good resistance against softening are needed

Products / shapes:

Round rods, wire coils, rectangular bars and solid profiles in age-hardened temper.

Softening behaviour – resistance against softening:

Room temperature hardness is presented in the following figure as a function of annealing temperature. Material at hard or aged temper.



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Chemical composition and corresponding standards:

Luvata Pori Oy alloy	Composition %	EN – CEN/TS 13388:2008	ASTM / USA
KrK101	Cr 0,5 – 1,0 % Zr 0,05 – 0,15 %	CuCrZr / CW106C	CDA C18150

Physical properties:

Density kg/dm³	Coefficient of linear expansion 1/K	Specific heat J/(kg x K)	Melting temperature °C
8,89	0,0000176	385	1075

Mechanical properties – round rods:

	Round dia < 10 mm	Round dia 10 < x < 25 mm	Round dia > 25 mm
Hardness HV	150 – 180 HV	130 – 170 HV	Approx. 130 HV
Tensile strength	450 – 600 N/mm²	400 – 600 N/mm²	Approx. 450 N/mm²
0,2% yield strength	400 – 550 N/mm²	350 – 550 N/mm²	Approx. 350 N/mm²
Elongation	10 – 20 %	10 – 20 %	Approx. 15 %

Electrical and thermal properties – typical values:

Electrical conductivity	vol	% IACS *	approx 78,0
	mass	%IACS	approx 77,5
	MS/m		approx 45,0
Electrical resistivity	vol	Ω mm²/m	approx 0,022
	mass	Ω g/m²	approx 0,198
Thermal conductivity (20 °C)	W / Km		320

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

Joining and machining:

Machinability rating (free cutting brass = 100)	Soldering	Brazing	TIG	MIG	EBW
20	Good	Good	NOT Recommended	NOT Recommended	NOT Recommended

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